

Menglin (Miley) Liu

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Summary

AI research scientist with a Ph.D. and 5+ years of experience building end-to-end NLP/LLM pipelines, multi-agent systems, and AI safety evaluation frameworks. Published first-author work on LLM prompt optimization and jailbreak attack mitigation. Creator of PoliPrompt (PyPI), an open-source prompt optimization toolkit. UNICEF National AI Expert deploying autonomous research agents at scale. Lead scientist for China's first AI Agent Safety Evaluation Standard.

Experience

Assistant Professor, Computational Social Science & AI (by courtesy) *The Chinese University of Hong Kong, Shenzhen* Sept 2025 – Present

- Founded the Reinforcement Learning & Responsible AI Lab; advise research on RLHF, red-teaming, and safe agent deployment.
- Built multimodal audio-text pipelines (speaker diarization, ASR, LLM inference) processing 1,700+ transcripts and 180k+ minutes of audio at scale.
- Collaborate with ByteDance on [OpenViking](#), an open-source context memory system for long-horizon AI agents; co-design retrieval and state-tracking modules.

National AI Expert *The United Nations Children's Fund (UNICEF)* April 2026 – Present

- Architect and deploy a suite of autonomous AI agents — Deep Research Agent, Document Coding Agent, and Data Agent — for large-scale evidence retrieval, multilingual document classification, and policy gap analysis.
- Implement uncertainty-aware annotation with calibrated confidence scoring and human-in-the-loop validation, reducing manual review burden by 70%+.

Lead Scientist, AI Industry Standards *Shenzhen AI Industry Association (SAIIA)* 2026

- Lead-authored the *Specification for Security Evaluation of AI Agents* (T/AIIA), defining adversarial testing protocols, risk grading, and guardrails across five dimensions: input security, behavioral safety, tool invocation control, output compliance, and data privacy.

Lead Research Scientist *SurveyFluency* Sept 2025 – Present

- Architect LLM-powered survey analytics stack: RAG-based semantic search over 10k+ survey items, automated survey ingestion via structured extraction, and synthetic respondent generation using persona-conditioned language models.

Selected Projects

PoliPrompt: Prompt Optimization Framework  2024

- Built and published Python package for LLM-based text classification featuring automatic prompt optimization via iterative refinement, dynamic few-shot exemplar selection, and calibrated uncertainty estimation. Reduces annotation cost by 80%+ vs. manual labeling with comparable accuracy.

Multimodal Agent Benchmark and Validator 2025 – Present

- Design benchmark datasets and evaluation harness for multimodal agent reasoning across text, audio, and image inputs; integrate CLIP and vision-language model pipelines for zero-shot visual grounding and cross-modal retrieval.

Education

Ph.D., Political Science – University of California, Davis 2019 – 2025

B.A., English Literature – Beijing Foreign Studies University 2015 – 2019

Selected Publications

- **Liu, M.** and G. Shi. *PoliPrompt: A High-Performance Cost-Effective LLM-Based Text Classification Framework*. [arXiv 2024](#) . Under review.
- G. Shi, J. Yin, D. Xie, Y. Li, and **Liu, M.** *A Unified Framework to Optimize Jailbreak Attacks with Prompt Tool Libraries*. Submitted to ICML 2026.
- **Liu, M.** and J. Wang. *Understanding Local Housing Debates through Automated Audio Analysis Using Diarization, Transcription, and LLMs*. In progress.
- **Liu, M.** and A. Liu. (2026). *Homeownership and Preferences for Government Redistribution in China*. [Cities](#) .

Technical Skills

Languages: Python, R, SQL **Cloud / Infra:** AWS (EC2/S3/Lambda), Docker, CI/CD

ML / DL: PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, PEFT/LoRA, RLHF

LLM / NLP: Prompt engineering, RAG, LangChain, function calling, agent orchestration, SpaCy, NLTK

Safety: Red-teaming, jailbreak evaluation, guardrail design, adversarial robustness testing

MLOps: Weights & Biases, MLflow, DVC, Git **Human Languages:** English (fluent), Mandarin (native)